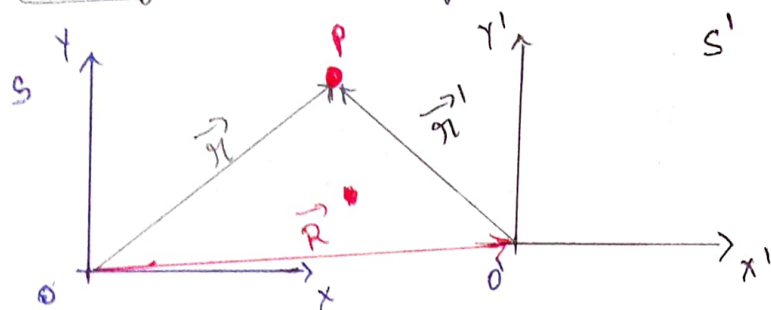


→ Moving frame of Reference



$\vec{r}' = \vec{r} - \vec{R}$

Relationship between position vectors in S & S' frame of reference. S' frame of reference is moving with a constant velocity $\vec{v}$. S is a frame of reference that is at rest.

- $\vec{R}$ is the position vector of S' frame from S frame.
- $\vec{r}$ is the position vector of "P" in S frame.
- $\vec{r}'$ is the position vector of "P" in S' frame.

→ The relative velocity between the two frames is $\vec{v} = \frac{d\vec{R}}{dt}$

→ for S' to be an inertial frame of reference $\frac{d\vec{v}}{dt} = 0$ } no acceleration.

$$\therefore \vec{r}' = \vec{r} - \vec{R}$$

$$\frac{d\vec{r}'}{dt} = \frac{d\vec{r}}{dt} - \frac{d\vec{R}}{dt}$$

$\vec{v}' = \vec{v} - \vec{v}$

relationship between velocities for particle "P" in S' & S frame of reference.

$$\therefore \vec{v}' = \vec{v} - \vec{v}$$

(2)

$$\frac{d\vec{v}'}{dt} = \frac{d\vec{v}}{dt} - \frac{d\vec{v}}{dt}$$

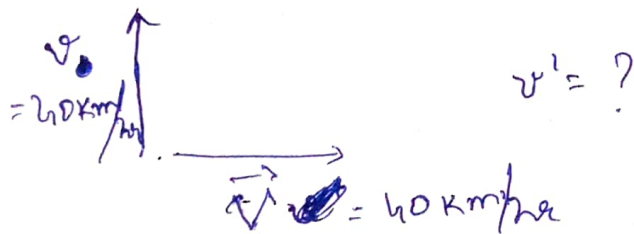
$$\therefore \vec{a}' = \vec{a} \quad \left\{ \text{for inertial frame of reference} \right.$$

$$\Rightarrow m\vec{a}' = m\vec{a}$$

$$\Rightarrow \vec{F}' = \vec{F} \quad \left\{ \text{Conservation of force.} \right.$$

- 1) A bird is flying at 40 km/hr speed towards north. A train travelling east is moving at 40 km/hr. Find the velocity of the bird WRT the passenger in the train.

Sol



$$\vec{v}' = \vec{v} - \vec{v} \quad \vec{v} = 40\hat{j} \quad v = 40\hat{i}$$

$$\vec{v}' = \vec{v} - \vec{v} = 40\hat{j} - 40\hat{i}$$

$$\vec{v}' = -40\hat{i} + 40\hat{j}$$

$$\therefore |\vec{v}'| = \sqrt{40^2 + 40^2} = 40\sqrt{2} = 56.6 \text{ km/hr}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(-1) = 45^\circ \text{ North of West}$$

